Mechanisms underlying movement patterns of the South American tortoise Chelonoidis chilensis

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Abstract

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Introduction

Animal trajectories are the result of a complex interplay between internal and external factors, including the animal's physiological state, environmental conditions, and interactions with other individuals. Understanding the mechanisms underlying movement patterns is crucial for predicting animal behavior, managing wildlife populations, and conserving biodiversity. How animals take decisions about when and where to move is a fundamental question in ecology and behavioral biology. Movement patterns can be influenced by a variety of factors, including resource availability, predation risk, social interactions, and environmental conditions. In this study, we focus on the South American tortoise Chelonoidis chilensis, a species that is classified as endangered. As a reptile, it is ectothermic and relies on external sources of heat to regulate its body temperature. This makes it particularly sensitive to environmental conditions, which can influence its movement patterns and behavior. By performing comuter simulations is possible to explore the effects of different factors on movement patterns and to identify candidate mechanisms that drive these patterns. Previous studies showed that tortoises movement is subdiffusive which suggests....

By analyzing GPS tracking data collected over several camàigns, we aim to uncover the statistical properties of their movement patterns and identify potential factors influencing their behavior.

Methodology

The study was carried out in a site of approximately 25 hectares, located about 20 km north of the city of San Antonio Oeste, Province of Río Negro, Argentina, where there is a stable population of C. chilensis with a large number of specimens. The area in general belongs to the Monte Austral phytogeographic unit, characterized by a shrub steppe with a predominance of Larrea spp. with several strata, the lower one being grasses and herbs, all with very little coverage, and particularly with very few cacti (Oyarzabal et al. 2018). The specific study area is characterized by a vegetation cover, mostly with xerophytic characteristics, with a predominance of grass clumps (Poaceae) and shrubs (León et al. 1998, Morello et al. 2012). In turn, the grass stratum has a higher species richness than the other Zygophyllaceae of the Monte steppes. The soil is rather sandy with a low percentage of clay and abundant pebbles. Close to the border of the study area, there is the Gran Bajo del Gualicho, a Patagonian plain of up to 70 meters below sea level, characterized by a high saline concentration. This is why there are more saline sectors, where we can find halophyte species such as Suaeda divaricata and several species of the genus Atriplex spp. Scarce precipitations give an average of approximately 255 mm in the year, with minor peaks in spring and fall. Annual average temperature is around 14.5 °C with significant daily thermal amplitudes and a persistence of the west and southwest winds, making the region more arid (Godagnone and Bran, 2008). The minimum average temperature is 1°C, and the maximum is 30 °C, with extreme recorded temperatures of -11.5 and 44.6 °C (data for 1961–2021 provided by the National Meteorological Service of Argentina). We tracked six tortoises continuously during a whole year using GPS data loggers (i-gotU GT-120 Mobile Action, Taiwan), recording geographical coordinates every 15 minutes. The positions were recorded continuously from 6am to 10pm and from 2am to 3am. The equipment was attached to the carapace of each individual using a camouflage tape (@Duck Tape, Real Tree Hardwood Camouflage) to mimic the environment, with the precaution of not leaving any sharp corners, in order to prevent the tortoise from getting trapped in the vegetation. After attaching the device, tortoises were released at the same place that were found. The weight of the equipment never exceeded

10In this work we present analysis and results spanning a whole year, involving a total of 20149 hours of GPS data loggers recording. Those recorded geographical positions corresponded to 813 trajectories from 6 individuals. The GPS data loggers were charged every fifteen days except during the brumation period, when animals were not disturbed and were observed every two weeks to check if they remained brumating at the same location. All recorded trajectories were preprocessed to exclude those corresponding to speeds above a threshold of 15 m/min, assessed as a reasonable maximum speed for the species, according to the speed empirical distribution presented in Fig.S3.

Results and Discussion

The distance traveled as a function of time was calculated from the (latitude, longitude) data by determining the distance on the Earth's surface using the Haversine function.

The total accumulated distance traveled was then calculated as:

$$\text{distot}(t) = \text{distot}(t-1) + \text{distance}(t)$$

Figure 1 shows all the measured quantities for turtle T10. There are some temporal gaps corresponding to missing data.

A similar behavior is observed for the other turtles (T6, T12, T30, T54, T79, T128, T185, and T206).

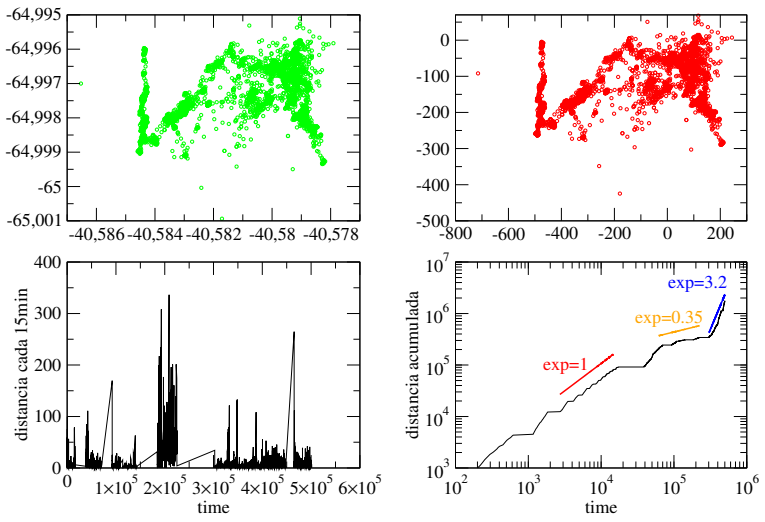


Figure 1: Datos de la Tortuga T10: A) Datos crudos de (lat,long) ; B) Lo mismo que en A) pero metros, para chequear que el pasaje de datos funcione. C) Distancia recorrida (cada 15min) en funcion del tiempo; D) Distancia acumulada vs time: los colores indican los exponentes del ajuste en cada periodo.

The distances traveled by the turtles were measured every 15 minutes. Based on the observed behavior, we can define an exponent as:

$$\text{P(d)} \approx d^{-\beta} \tag{1}$$

The mean exponent for the 9 turtles is: $\beta \approx 2.1(2)$, as shown in Figure 2.

La nueva figura de MSD solo con los datos de hora solar (9 a 17hs) Figure 4.

Mean square deviation (MSD)

ESTO YA NO ES CIERTO: By calculating the MSD for each tortoise, markedly different behaviors are observed, both among tortoises and depending on the time period (Figure 4). We can fit an exponent:

$$\text{MSD} \approx t^{\alpha} \tag{2}$$

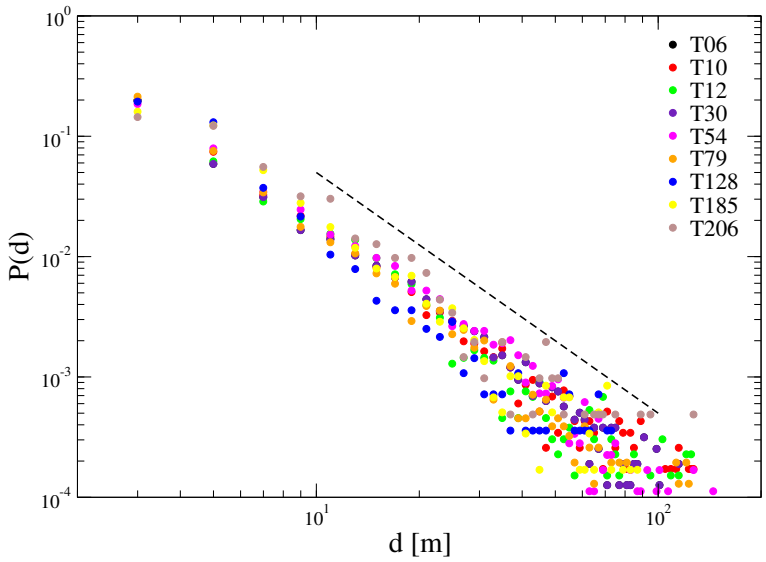


Figure 2: Distribucion de distancias recorridas cada $\Delta t = 15min$, tomando sólo hora solar.

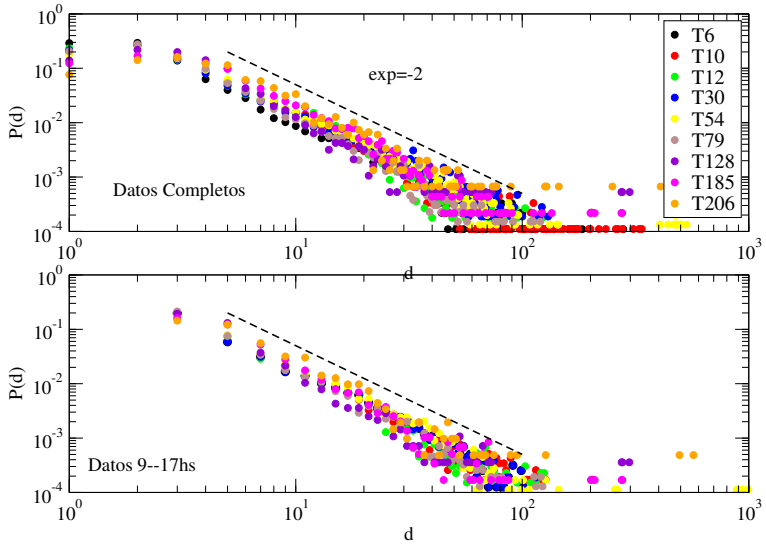


Figure 3: Distribucion de distancias recorridas cada $\Delta t = 15min$.

Seasonality

If we attempt to separate the behavior according to biological/climatic periods, namely: November–January, February–March, and April–October, we observe that the MSD behavior is **NOT** markedly different, with exponents ranging from subdiffusive to diffusive regimes (see Figure 4). The seasonal periods still need to be refined. Here, only the calendar months were considered, but factors such as temperature, late spring conditions, and related environmental variables should probably also be taken into account.

Para todas las tortugas el MSD Figure 4 y la distribucion de distancias $P(d)$ Figure 5

In both figures, Fig. **??** and Fig. 5, the dotted lines correspond to the best fits obtained. Table 1 lists the values of these exponents ($MSD(t) \approx t^{\alpha}$ and $P(d) \approx d^{-\beta}$). As can be seen, there is considerable dispersion among turtles and across different periods of the year.

It should be noted that for some tortoises (T128, T185, and T206) only limited data are available. For example, no data are available for the November–January period, and for the April–October period only data from the first months (April and May) are available, as indicated in Figures **??** and 5.

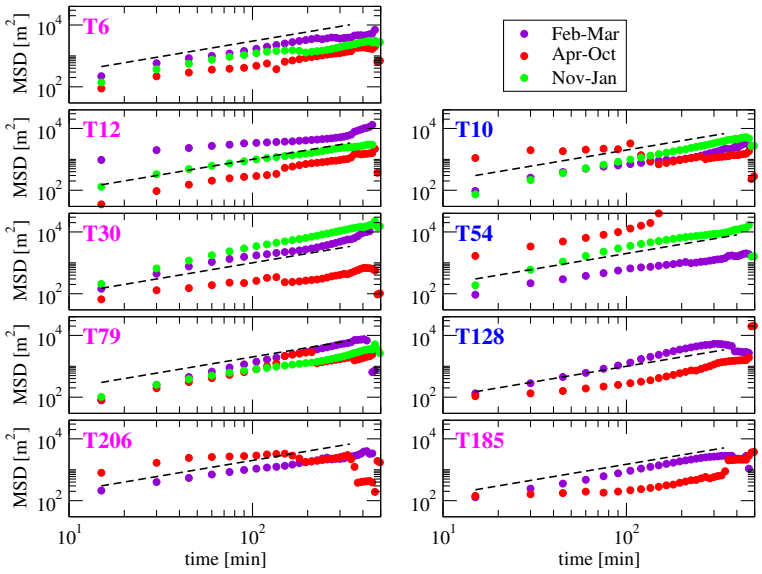


Figure 4: MSD versus time for all turtles (labeled as *Tnumber*) recorded during daylight hours (9:00 a.m. to 5:00 p.m.), categorized by seasonality as indicated. The color of each label indicates the turtle's sex (magenta for females and blue for males). Dotted black lines represent a slope corresponding to diffusive behavior ($\alpha \approx t^1$). Further details are given in the text.

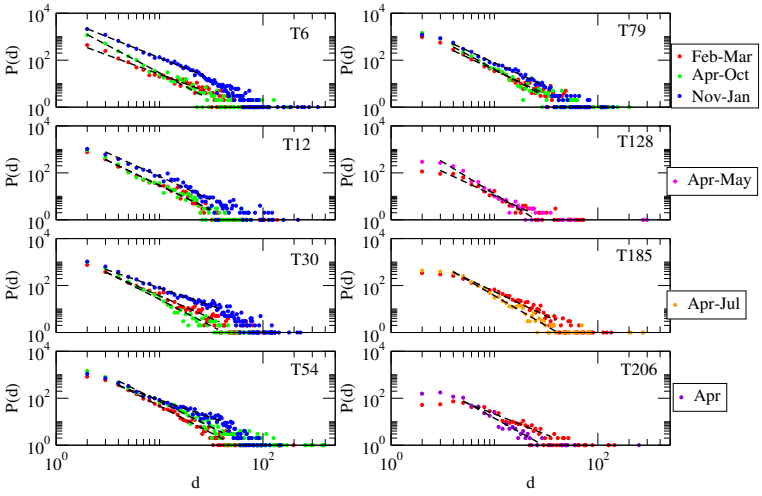


Figure 5: Distribucion de distancias recorridas en intervalos de 15min segun separacion estacional para todas las tortugas

Males and females

So far, the MSD appears to exhibit a power-law behavior during the February–March period. Beyond this interval, it becomes difficult to obtain reliable fits.

Within this period, it may be useful to separate the data between males and females in order to investigate possible differences.

Figure 7 shows the MSD behavior separated by males and females. The exponent α (Eq. 2) does not appear to change significantly between males and females; rather, there is considerable dispersion in the data (see Table 2).

A similar situation is observed for the distribution of traveled distances (Figure 8), although in this case the dispersion in the curves may be larger among females than among males.

Table 2 lists the values of the exponents.

Exponente alfa (MSD)				Exponente beta (P(d))			
Tortuga	Feb-Mar	Abril-Oct	Nov-Ene	Tortuga	Feb-Mar	Abril-Oct	Nov-Ene
T6	0.269	0.915	0.565	T6	1.62	2.3	1.76
T10	0.064	0.539	0.883	T10	1.8	2	2.4
T12	0.34	0.67	0.262	T12	2.1	2.1	1.99
T30	0.774	-0.117	0.708	T30	1.92	2.2	1.51
T54	0.901	0.001	0.629	T54	2.29	2.2	1.5
T79	0.364	0.075	0.292	T79	2.08	2.4	2.01
T128	0.588	0.865	S/D	T128	2.1	2.7	
T185	0.651	0.367	S/D	T185	2.1	2.5	
T206	0.792	0.54	S/D	T206	1.75	2.4	
Valor medio	0.5(3)	0.4(4)	0.5(2)	Valor medio	2.0(2)	2.3(2)	1.9(3)

Figure 6: Table 1: Exponents of MSD and $P(d)$ for each turtle and for each period. Males and females are distinguished by color, and the exponents were calculated using all available data.

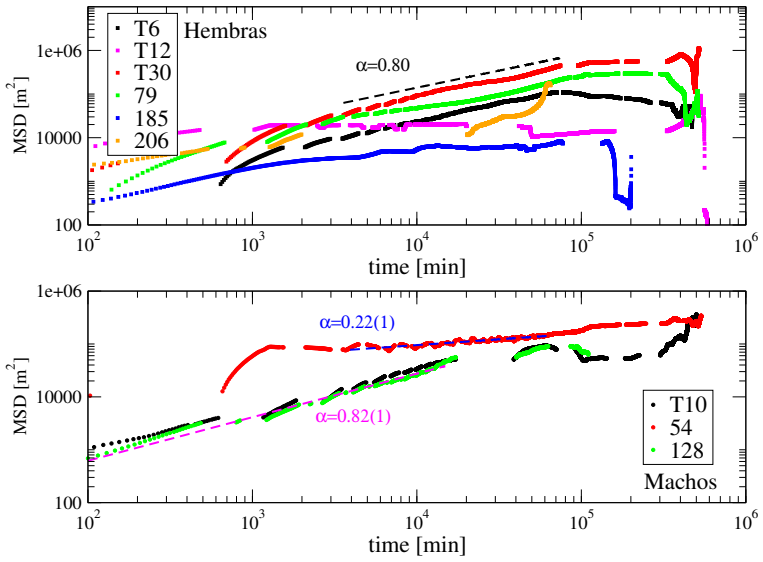


Figure 7: MSD separated in males and females

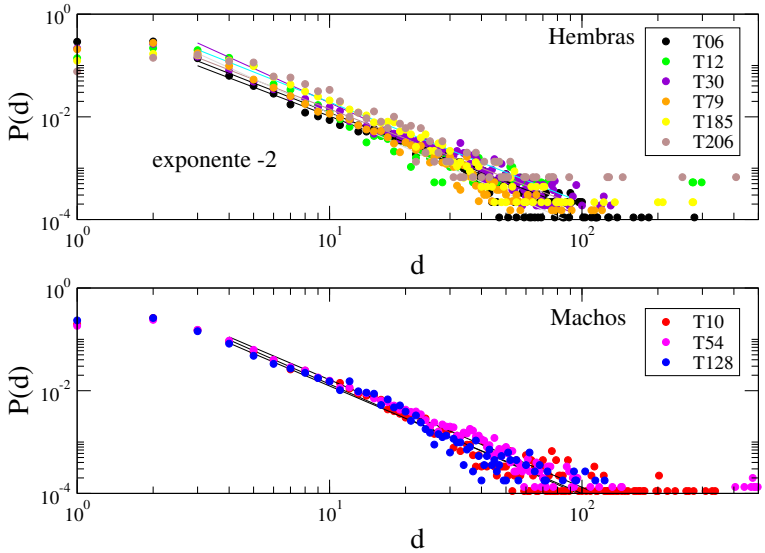


Figure 8: P(d) vs d separated in males and females

Bouts of walking behaviour

We have shown that the power-law distribution of displacements can be explained by assuming the tortoise performs random walks with constant speed interspersed with resting periods, provided the duration of the walking bouts

		α (MSD)	β (P(d))
Hembras	T6	0.83	1.88
	T12		1.92
	T30	0.78	1.91
	T79	0.74	2.22
	T185	0.79	2.22
	T206	0.53	1.95
Machos	T10	0.82	2.16
	T54	0.22	2.09
	T128	0.82	2.09

Figure 9: Table 2: MSD and $P(d)$ exponents for each tortoise, separated by males and females, calculated using all available data without separation by periods.

is distributed with a power-law, or Pareto, distribution,

$$P_b(t) = \frac{\alpha - 1}{t^\alpha}, \qquad t \geq 1,$$

where $P_b(t)\,dt$ is the probability that a bout of walking will last from time $t = 0$ to a time between t and $t + dt$.

This behaviour can also be explained by considering the transition probability from walking to resting, $\gamma(t)$. The transition probability is the probability that the tortoise will stop walking at a time between t and $t + dt$, given that it is walking at time t, i.e.

$$\gamma(t) = \frac{P_b(t)}{\text{P. of walking at time } t} = \frac{P_b(t)}{1 - \text{P. of having stopped between 0 and } t} = \frac{P_b(t)}{1 - \int_0^t P_b(t')\,dt}.$$

Since $\int_1^t (\alpha - 1)t^{-\alpha} = 1 - t^{1-\alpha}$, we obtain

$$\gamma(t) = \frac{\alpha - 1}{t}, t \geq 1,$$

i.e.\ the probability that the tortoise will decide to rest gets smaller the longer it has been walking.

Power-law-distributed bouts of behaviour have been discussed before [**?**], [**?**], but not everyone seems to agree on power laws [**?**].

Transition rate on a fixed maximum behaviour time

It has been shown that tortoises if can be clasified into two groups, in which both follows a power law distributions, $f(t)$, of exponent α with a maximum time of behaviour, t_{max}, [**?**]. In order to facilitate the theoretical analysis, we can infer a transition rate from one state to another, which is a function of the time spent in the current state, t. Another thing to consider is that in our simulation we have discrete time steps, δt. We will obtain the transition rate, through the following relation:

$$\gamma(t) = 1 - P(f(t + \delta t)|f(t)) \tag{3}$$

Where $P(f(t + \delta t)|f(t))$ is the conditional probability of being in the same state at time $t + \delta t$ given that it was in that state at time t. This conditional probability can be expressed as:

$$P(f(t + \delta t)|f(t)) = P(f(t)|f(t + \delta t))\frac{P(f(t + \delta t))}{P(f(t))} \tag{4}$$

in which $P(f(t)|f(t + \delta t)) = 1$ and $f(t) = At^{-\alpha}$ with A a normalization constant. $P(f(t))$ can be calculates as follows:

$$P(f(t)) = \int_t^{t_{\max}} f(t')\,dt' = \frac{A * \left(t_{\max}^{1-\alpha} - t^{1-\alpha}\right)}{1 - \alpha} \tag{5}$$

in this equation we are assuming that the state began at time $t = 0$ so if we apply this to the conditional probability we have:

$$P(f(t+\delta t)|f(t)) = \frac{t_{\max}^{1-\alpha} - (t+\delta t)^{1-\alpha}}{t_{\max}^{1-\alpha} - t^{1-\alpha}} \tag{6}$$

Finally, we can express the transition rate as:

$$\gamma(t) = \frac{(t+\delta t)^{1-\alpha} - t^{1-\alpha}}{t_{\max}^{1-\alpha} - t^{1-\alpha}} \tag{7}$$

1 Model

The model developed in this work describes the movement of each tortoise as a *Stop and Go* random walker moving in a homogeneous continuous space, that is, without refuges or any other structures that introduce spatial heterogeneity. We simulate $N \approx 100.000$ individuals independently, with no interaction between them, each alternating between two states: movement (*Go*) and rest (*Stop*). The day-night cycle is incorporated solely as an active time window, analogous to the one used during GPS data acquisition (Section **??**): outside this interval, which coincides with the nocturnal resting hours of the species, the individual remains motionless. Each simulation spans two months.

There are at least two possible strategies to determine how long an individual remains in each of the two states. The first consists of defining an instantaneous transition rate $\gamma(t)$, a function of the time already spent in the current state, and evaluating it at every discrete time step δt of the simulation to decide whether the individual switches state. Although this approach naturally accommodates a time-dependent rate, it requires evaluating a transition probability at every time step of each bout, which considerably increases the computational cost when simulating a large number of individuals.

The second strategy, which we adopt in this work, consists of drawing the total time to be spent in a state directly upon entering it from its full duration distribution. To this end, we use inverse transform sampling (*inverse-CDF sampling*): given a probability density $f(t) \propto t^{-\alpha}$ truncated between a minimum time t_{min} and a maximum time t_{max}, its cumulative distribution function can be inverted to obtain a closed-form expression that, given a random number u uniformly distributed in $[0, 1]$, directly yields the sampled time:

$$t = \left[(t_{max}^{1-\alpha} - t_{min}^{1-\alpha})\, u + t_{min}^{1-\alpha} \right]^{\frac{1}{1-\alpha}} \tag{8}$$

This method yields, in a single operation, the full time that an individual will spend in the movement or rest state, without having to evaluate the transition probability step by step. Since we simulate a large number of individuals over long time spans, this reduction in computational cost is crucial for the feasibility of this work.

2 Stop and Go dynamics

In the *Go* state, the individual moves at discrete time intervals δt, taking steps of constant length L. The direction of each step is not drawn independently; instead, it follows a correlated random walk: the new orientation angle θ_{i+1} is obtained from the angle of the previous step, θ_i, by adding Gaussian noise with zero mean and standard deviation σ_θ,

$$\theta_{i+1} = \theta_i + \eta_i, \qquad \eta_i \sim \mathcal{N}(0, \sigma_\theta^2). \tag{9}$$

The position of the individual is therefore updated at each time step according to

$$x_{i+1} = x_i + L\cos(\theta_{i+1}), \qquad y_{i+1} = y_i + L\sin(\theta_{i+1}). \tag{10}$$

This angular correlation introduces directional persistence into the trajectory: unlike a simple random walk, in which the direction is drawn independently at each step, the individual tends to maintain its heading over short time scales, producing locally straight trajectories that curve gradually as angular perturbations accumulate.

In the *Stop* state, by contrast, the position of the individual remains fixed, $x_{i+1} = x_i$ and $y_{i+1} = y_i$, and the only process that advances is the count of the time elapsed in that state.

The duration of each bout, both *Go* and *Stop*, is drawn upon entering the corresponding state using the inverse transform sampling method described above, from the truncated power-law distribution associated with each of the two states.

3 Computational Results

3.1 First configuration: constant walking time

Figure 10 shows several displacement distributions, with $\sigma_\theta = 0.2$ and a constant walking time $T_{\text{walk}} = 10$ t.s. (time steps, i.e., 10 simulation steps) for all three curves. What varies between the curves is the exponent of the rest-time distribution.

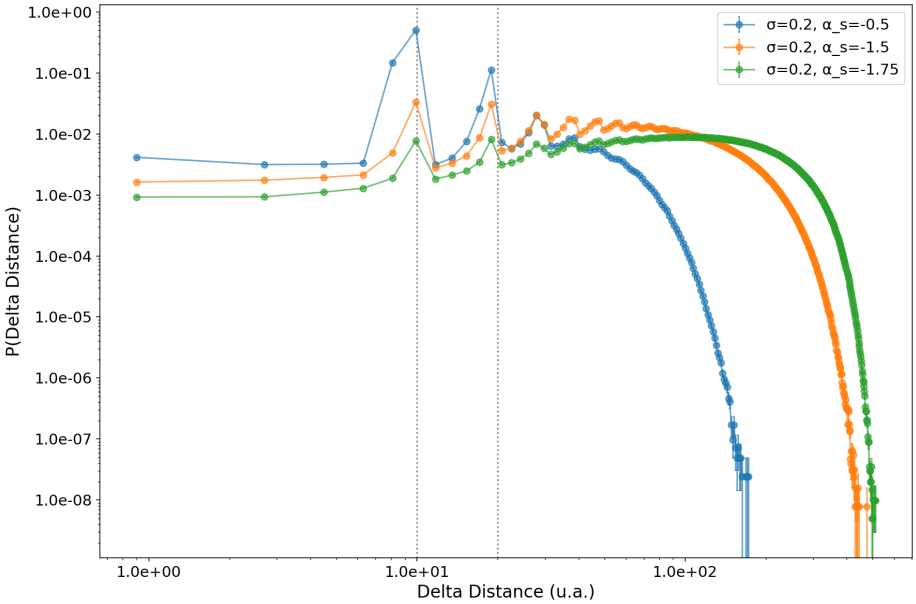


Figure 10: Displacement distribution for the first model, with constant walking time.

In this figure, we observe that the displacement distributions exhibit characteristic distances that are approximately multiples of $L\,T_{\text{walk}}$. Because the model adds Gaussian noise to the direction of motion, these characteristic values lie slightly below the aforementioned multiples, since the exact value is only attained when the individual moves in a straight line. These peaks correspond to a given number of walking bouts completed within the 15-minute window; for example, the first peak corresponds to a single walking bout, after which the individual remained at rest for the remainder of that window. As a consequence, these characteristic distances become blurred as the rest-time exponent α_S increases, since long rest times become less likely.

Given the discrepancy between the distributions obtained and the empirical distribution, we conclude that this model cannot reproduce the behavior observed in the tortoises, and we therefore discard it.

3.2 Second configuration: constant rest time

In this second configuration, we keep the rest time constant and let the walking time follow a power law. Figure 11 shows the displacement distribution for different rest times, keeping the walking-time exponent and σ_θ fixed.

The key feature of this figure is that, unlike the first model, the distributions display power-law behavior, and they converge to the same distribution once the rest times exceed the duration of the analyzed window (15 minutes). This configuration reproduces the power-law behavior with exponent $\alpha_d \approx 2$ observed in the tortoises for rest times longer than 500 seconds. Having achieved this, we analyze the MSD of this model, shown in Figure 12.

The figure shows that the MSD of all the simulations studied is initially superdiffusive and crosses over to diffusive behavior at long times.

3.3 Third configuration: power-law distributions for both states

The last configuration we tested is one in which both time distributions follow a power law, $f_S(t) \propto t^{-\alpha_S}$ for rest and $f_W(t) \propto t^{-\alpha_W}$ for walking, with $\alpha_S, \alpha_W > 0$. To explore the parameter space, we performed a sweep over $\alpha_S \in [0.5, 2.0]$, $\alpha_W \in [1.0, 3.0]$ and $\sigma_\theta \in [0.2, 2.0]$, computing in each case the 15-minute displacement distribution, $P(d) \propto d^{-\alpha_d}$. As an example, Figure 13 shows these distributions for different values of α_S, with $\alpha_W = 1.75$ and

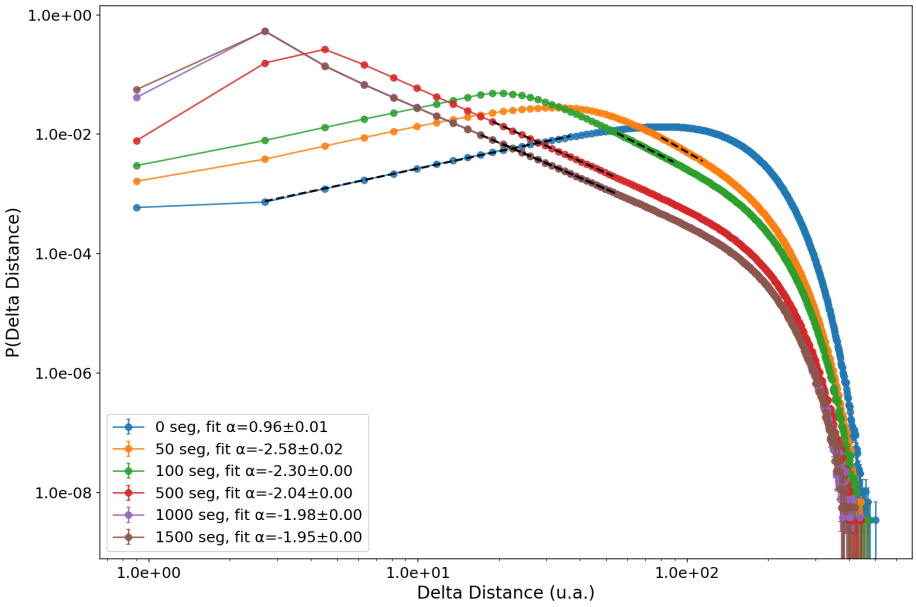


Figure 11: Displacement distribution for the second model, with constant rest time.

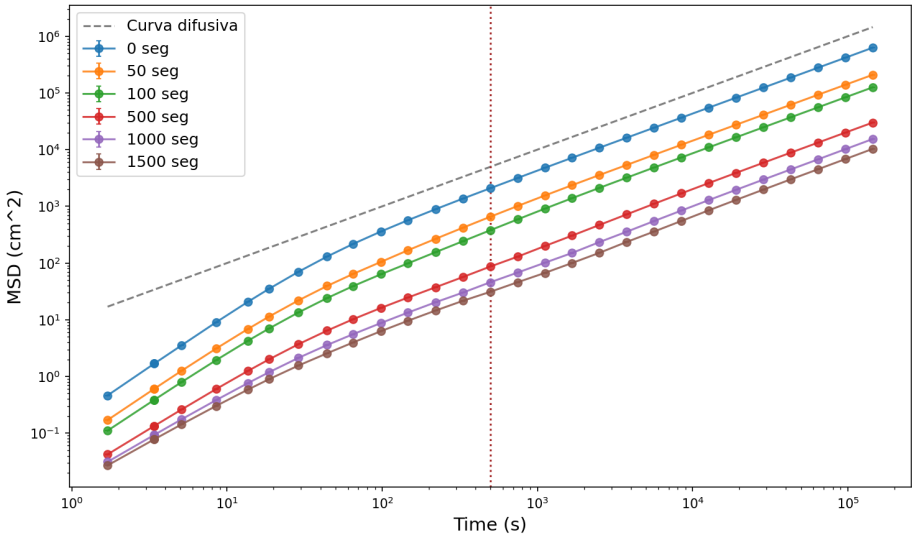


Figure 12: MSD for the second model, with constant rest time.

$\sigma_\theta = 0.2$ held fixed. Across the whole sweep, the decreasing power-law regime ($\alpha_d > 0$) appears only when the rest distribution decays more slowly than the walking distribution, that is, when $\alpha_S < \alpha_W$.

Within the region $\alpha_S < \alpha_W$, we searched for combinations that reproduce the empirically observed behavior. Figure 14 shows the best configuration obtained, with $\sigma_\theta = 0.25$, $\alpha_W = 1.8$ and $\alpha_S = 1.1$, that reproduces the empirical exponent, $\alpha_d \approx 2$, between $d \in [20, 200]$ (u.a.).

As in the previous case, once the displacement distribution had been reproduced, we analyzed the MSD of this model, shown in Figure 15. The MSD displays ballistic behaviour at short time scales, with an exponent that transitions to $\beta = 1$ at long time scales.

This convergence to a difussive behaviour is mainly due to the maximum time for the power law distribution, which lets the difussive behaviour of the gaussian distribution of the angular variation to control the MSD functionality at longer times. Also the maximum distribution times allows for several switches between the states, if we had no maximum times, due to the exponents of the distribution we are analysing, we would have little movement of the tortoises. In the Figure 16, we show three examples of trajectories with the best exponents with this configuration.

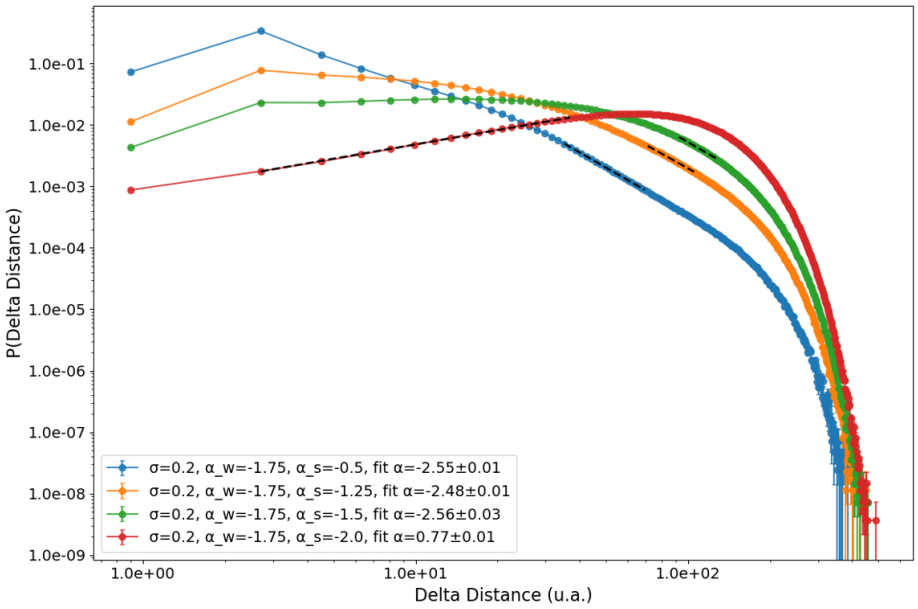


Figure 13: 15-minute displacement distribution for the third model, with $\sigma_\theta = 0.2$, $\alpha_W = 1.75$ and different values of α_S.

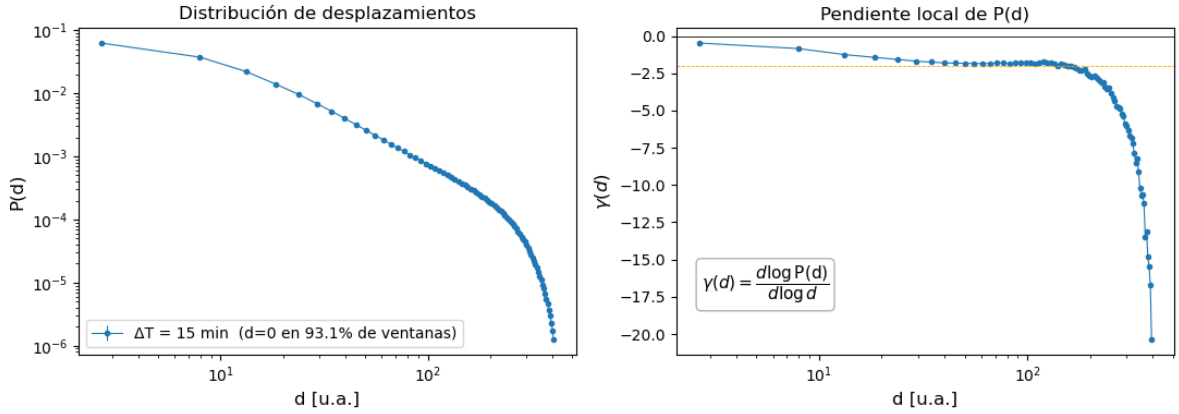


Figure 14: 15-minute displacement distribution for $\sigma_\theta = 0.25$, $\alpha_S = 1.1$ and $\alpha_W = 1.8$.

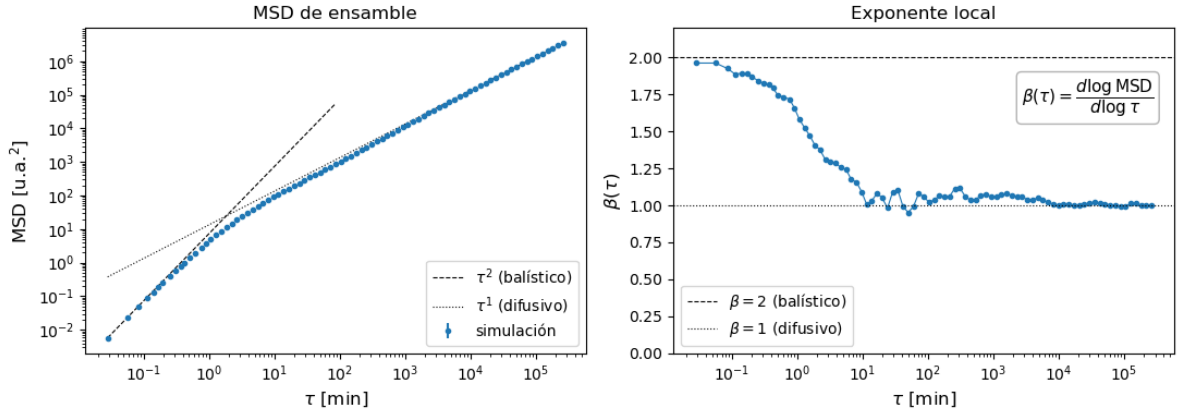


Figure 15: MSD for $\sigma_\theta = 0.25$, $\alpha_S = 1.1$ and $\alpha_W = 1.8$. The gray lines indicate balistic and diffusive behavior.

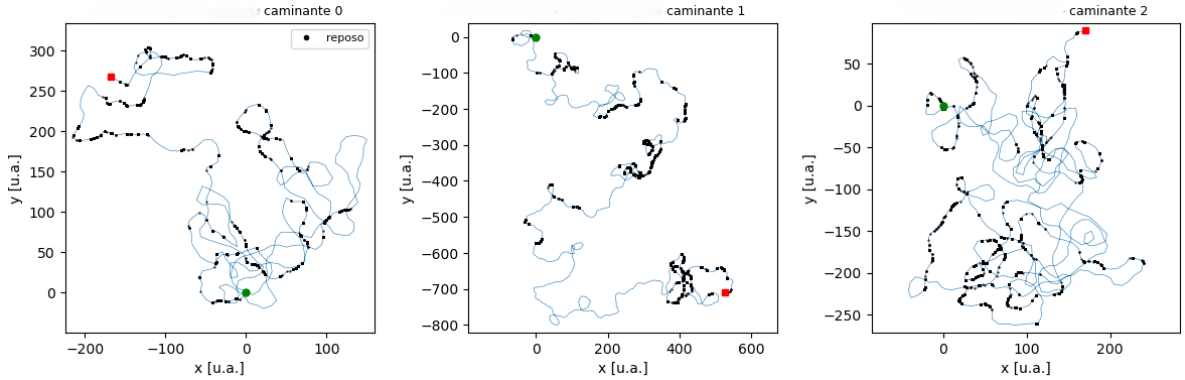


Figure 16: Three examples of trajectories for $\sigma_\theta = 0.25$, $\alpha_S = 1.1$ and $\alpha_W = 1.8$.

Supplement

By testing a different temporal interval, $\Delta t = 30\,\text{min}$, the exponent does not change, nor is any improvement observed in $P(d)$ (Figure 18).

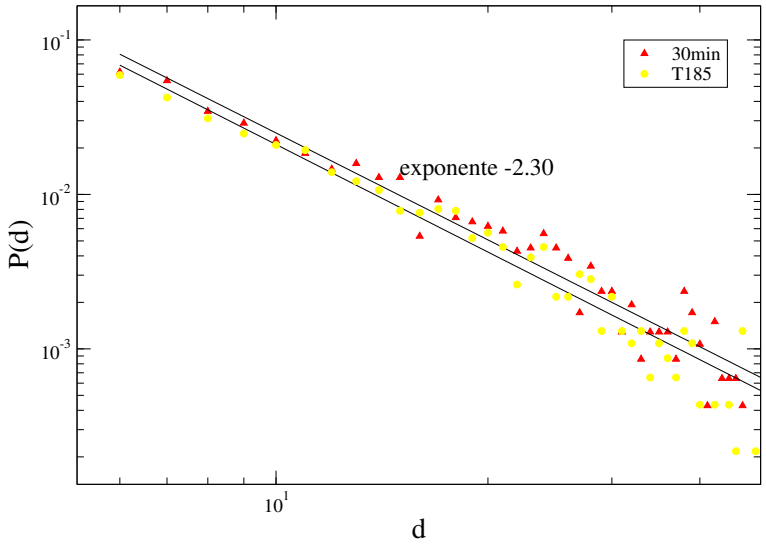


Figure 17: Distribution of traveled distances for T185, using $\Delta t = 15\,\text{min}$ and $\Delta t = 30\,\text{min}$.

Supplement

By testing a different temporal interval, $\Delta t = 30\,\text{min}$, the exponent does not change, nor is any improvement observed in $P(d)$ (Figure 18).

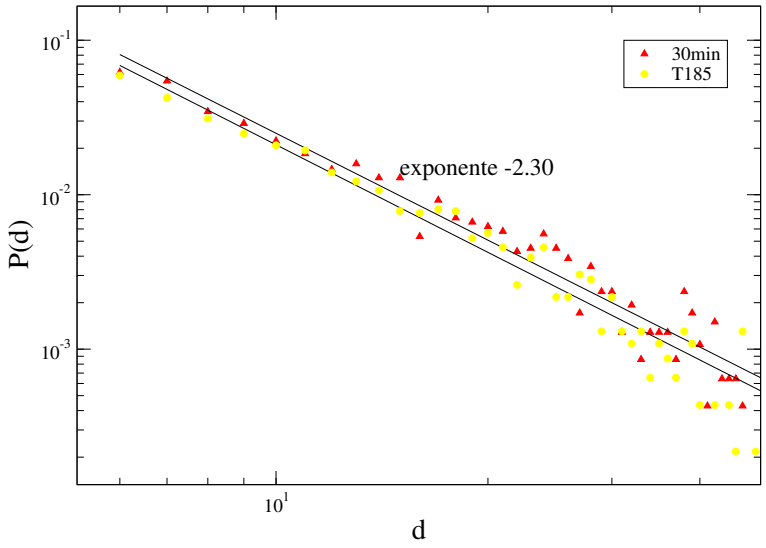


Figure 18: Distribution of traveled distances for T185, using $\Delta t = 15\,\text{min}$ and $\Delta t = 30\,\text{min}$.